ScamShield AI

Creating Intelligent Chatbots for Practical Applications

Student details	Assessment details
Imrul Kayes Shaikh Registration number: 1000700 Viraj International School	Career-related study: Artificial Intelligence Course: Artificial Intelligence: Real-World Applications and Implications SA-1 | 60 marks

Project links

GitHub repository:
https://github.com/Imrul-loq/IDAI101-1000700-Imrul-Kayes-Shaikh

Live chatbot on Vercel:
https://idai101-1000700-imrul-kayes-shaikh.vercel.app/

The repository is private. Assessor access has not yet been confirmed. The website was deployed and a successful greeting was shown in the student-provided screenshot.

Problem and intended users

ScamShield AI is an educational scam-awareness assistant for India. It helps students, families and less experienced internet users describe suspicious requests, understand warning signs and consider a safer next step. Its intended value is clear guidance before a person acts under pressure.

The design assumes users need simple language, questions that collect missing details and an explanation of the chatbot's limits. These needs were inferred from the assignment scenario; no completed interviews or survey findings are claimed.

Scope and boundaries

The chatbot covers job offers, UPI requests, phishing, OTP requests, impersonation and remote-access concerns. It includes a guided check and a three-question quiz. It cannot verify a sender, prove fraud, inspect accounts, recover money or submit an official report. Users are asked to use fictional examples and avoid private details.

Design and implementation

The prepared Dialogflow ES export contains **36 intents, 12 custom entities and 372 example training phrases** across 31 non-fallback intents. Five additional intents handle fallback. These configuration counts do not measure accuracy and are not a fresh audit of the live console.

Conversation design

Intent group	Purpose
Welcome, MainMenu, AboutLimits, Privacy	Introduce the assistant, choices and privacy boundaries.
GuidedCheck, ExplainCheck, NoExposure	Collect a request, explain caution using context and respond when nothing was shared.
FakeJobAdvice, OTPAdvice, PhishingAdvice, UPIReceiveSafety, QRSafety	Explain common scam-related situations.
ImpersonationAdvice, PressureAdvice, SafeVerification, SafetyTips	Explain claimed identities, pressure tactics and independent checks.
MoneyLostEmergency, SecretSharedEmergency, ClickedLink, RemoteAccessEmergency	Provide guidance for different reported exposure situations.
ReportingSteps, EvidenceChecklist, HumanHelp, CancelAndRestart	Support reporting, human help and conversation recovery.
QuizStart; Quiz1/2/3 Safe and Unsafe intents; fallback intents	Deliver quiz feedback and keep unmatched input within the appropriate conversation stage.

Entities and slot filling

The 12 custom entities are scam_type, contact_channel, requested_action, claimed_identity, sensitive_item, payment_method, link_type, pressure_tactic, job_offer_type, report_need, evidence_type and quiz_choice. They represent categories such as “WhatsApp” or “OTP”, rather than requesting actual secret values.

GuidedCheck requires the topic, contact channel and requested action. Example: **check a message** → **job offer** → **WhatsApp** → **send money**. It asks for missing values before returning a summary. The **review** context supports “Why be cautious”; **quiz1, quiz2 and quiz3** separate quiz stages. The README provides the detailed intent and entity inventory.

Website integration

A responsive HTML/CSS/JavaScript page embeds Google Dialogflow Messenger. Vercel serves the page, while Dialogflow ES handles intent matching and conversation context. No custom webhook, Python backend or GPT integration is used. The webpage requires no developer-supplied API key; Messenger communicates with Google's service. The page's prompt buttons copy text for the user to paste into the chat.

Evidence and reflection

Check	Available evidence and status
Deployment and greeting	Student screenshots show Vercel Ready, the live page, “Hi” and the ScamShield introduction. Demonstrated.
Guided request and follow-up	Earlier pasted Dialogflow replies collected job offer, WhatsApp and send money; the explanation retained those details. Demonstrated through reported console interaction.
Nothing shared	An earlier pasted reply addressed no reported exposure. Demonstrated through reported console interaction.
Five distinct scenarios	A complete five-scenario evidence set has not been recorded. The guided flow and its follow-ups are one workflow.
Quiz, fallback and emergency branches on Vercel	Not verified in the recorded live-site evidence. Further testing was deferred at the student's request.

Improvement and practical learning

Earlier attempts loaded the chat window without replies. A local-file screenshot showed a CORS error. The later Vercel screenshot showed a successful greeting. This records an improvement in the demonstrated interaction, but does not establish the root cause of every earlier failure or prove that all branches work.

The project shows how intents identify a user's purpose, entities capture useful details and contexts support follow-ups. A key learning is that a deployed page and a fully tested chatbot are different achievements. Clear limits and honest evidence are especially important for a safety-related assistant.

Documentation and submission status

GitHub includes index.html and a detailed README with the overview, live link, intent/entity explanation, integration and deployment instructions. The Dialogflow backup and supporting evidence still need to be added to the replacement repository. Assessor access also remains to be arranged. No overall accuracy score or full testing pass is claimed.

Privacy and assistance acknowledgement

Use made-up examples. Do not enter real OTPs, PINs, passwords, card details or identity documents. The page has no custom chat database, but Google and hosting providers may retain operational logs. ChatGPT assisted with the agent configuration, website and documentation. Results described here are limited to the evidence supplied during the project.

Sources used for this summary

1. Supplied SA-1 assignment brief and assessment rubric (60 marks).
2. Prepared ScamShield AI agent export and repository README.
3. Student-provided Dialogflow replies and Vercel deployment/greeting screenshots.

The separate ParentConnect example informed document organisation only; its implementation and results are not claimed for this project.